

Project Case	
COSC6050003 COSC6067052 Programming for Business	
Computer Science	E263-COSC6050-EV01-00
<i>Valid on Even Semester Year 2025/2026</i>	Revision 00

1. Mahasiswa tidak diperkenankan untuk:

Students are prohibited from:

- Berdiskusi dan/atau bekerja sama dengan mahasiswa lainnya,
Discussing and/or cooperating with other students,
- Melihat sebagian atau seluruh jawaban mahasiswa lainnya,
Seeing a part or the whole answer from other students,
- Membuka dan/atau menyalin jawaban dari buku, catatan, video, dan jenis referensi lainnya,
Open and/or copy answer from books, notes, videos, and other references,
- Membuka dan/atau menyalin jawaban dari internet,
Open and/or copy answer from the internet,
- Mengumpulkan jawaban yang tidak sesuai dengan tema soal,
Submitting an answer with a different theme from the given case,
- Melakukan tindakan yang menyebabkan jawaban dicontek oleh orang lain atau kelompok lain, baik disengaja maupun tidak disengaja,
Doing action that could result the answer being copied by someone or other groups, intentionally or unintentionally,
- Melakukan tindakan kecurangan lainnya.
Committing other dishonest actions.

2. Jika mahasiswa dan/atau terbukti melakukan tindakan seperti yang dicantumkan pada butir ke-1, maka nilai mahasiswa dan/atau kelompok yang melakukan kecurangan, baik menyontek atau dicontek, akan dinolkan sesuai dengan peraturan yang berlaku.

If it has been proven that a student and/or group has committed dishonest actions outlined in point 1 above, the students and/or groups related to the incident, regardless of which one copies or has their answer copied, will be issued a score of zero according to the regulation.

3. Jawaban yang dapat diterima dan dinilai adalah jawaban yang dikumpulkan sebelum batas waktu yang telah ditentukan.

The answer must be submitted before the designated deadline to be accepted and graded,

4. Jawaban akan dinilai berdasarkan teknik yang diajarkan dalam praktikum dengan menggunakan software yang telah ditentukan.

The scoring will be based on the materials taught during the practicum classes using the designated software. Using different software than requested may result in your answer not being graded.

5. Jika Anda tidak membaca peraturan ini, maka Anda dianggap sudah membaca dan menyetujuinya.

By taking this exam, you agree to these regulations, regardless of whether you have read it or not.

6. Persentase penilaian untuk matakuliah ini adalah sebagai berikut:

The score will be distributed as follows:

Tugas Mandiri <i>Assignment</i>	Proyek <i>Project</i>	UAP <i>Final Exam</i>
40%	60%	-

7. Perangkat lunak yang digunakan pada matakuliah ini adalah sebagai berikut:

This course uses the following software:

Software <i>Software</i>
Visual Studio Code XAMPP 8.2.12

8. Ekstensi file yang harus dikumpulkan untuk matakuliah ini adalah sebagai berikut:

Your answers must be in the following file extensions:

Tugas Mandiri <i>Assignment</i>	Proyek <i>Project</i>	UAP <i>Final Exam</i>
HTML, CSS, JS	PHP	-

Soal*Case***SEVeros Blackmarket**

SEVeros Blackmarket is an online marketplace where **users** can view and purchase weapons that are currently available on the market. Since this is a black-market environment, the items listed are typically illegal. All weapon listings, including their setting, price, and stock, are managed and updated by the **admin**, who controls the available inventory.

The system is built as a **PHP-based website** that handles all **Create, Read, Update, and Delete (CRUD)** operations through HTTP requests. It supports **three user roles: Admin, Member, and Guest** (non-logged-in user), each with **different pages** and **access permissions** based on their role.

- Guest
 - Home
 - Login
 - Register
- Member
 - Home
 - Marketplace
 - Arsenal
 - Transaction
- Admin
 - Dashboard
 - Manage Weapon
 - Manage Weapon Type
 - Manage Payment Method

Here is the entity relationship diagram of the **database** that you will be working with:

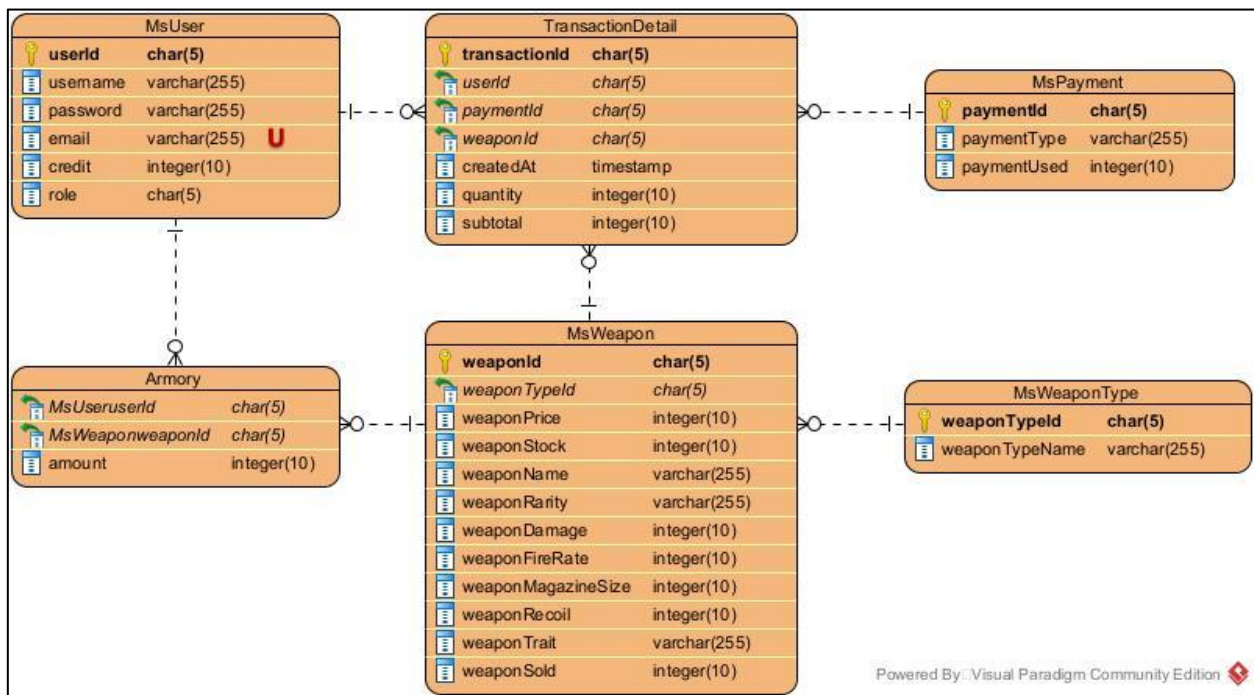


Figure 1. Entity Relationship Diagram

Each page shows **the name of Member or Admin**. If the member **hasn't logged in (guest)**, then the user **can access the login page** instead. You need to build this website based on the following criteria:

1. Home Page

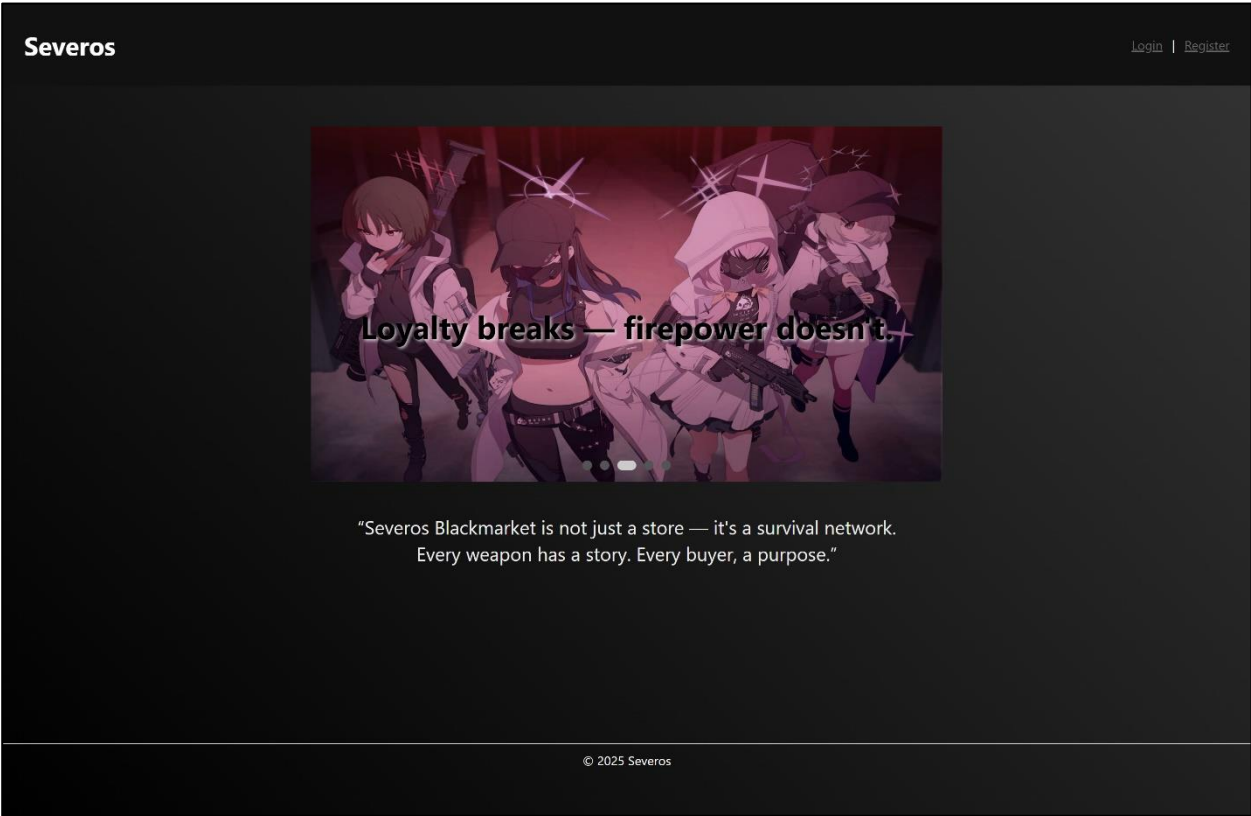


Figure 2. Home Page

- **Show** navigation bar with content according to user role status.
 - **Show** the company name.
 - **Show** link for accessible pages based on user **roles**.
 - **Show** username if authenticated.
 - **Add** login, register or logout button according to user authentication status.
 - **Clear** cookies after successful logout.



Figure 3. Guest Navigation Bar

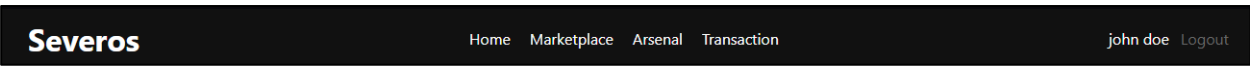


Figure 3. User Navigation Bar

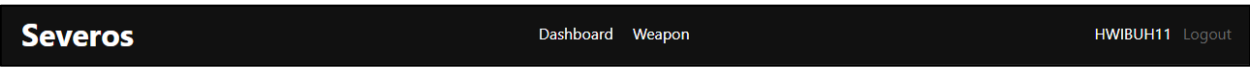


Figure 5. Admin Navigation Bar

- **Display** an image slider and a slogan for the company. You are free to use any image that is suitable for the website.
- **Display** footer. You are free to design the footer as long as it's suitable for the website.

2. Login Page

Figure 4. Login Page

- **Validate** that the input must not be empty. If not, **display** error message.
- **Validate** that the password length must be **greater than equal to 8 characters**.
- **Add** register button to redirect user to register page.
- **Create** cookies after successful login. Make sure each role **can't access** other roles page and authenticated user **can't access** login and register page.

3. Register Page

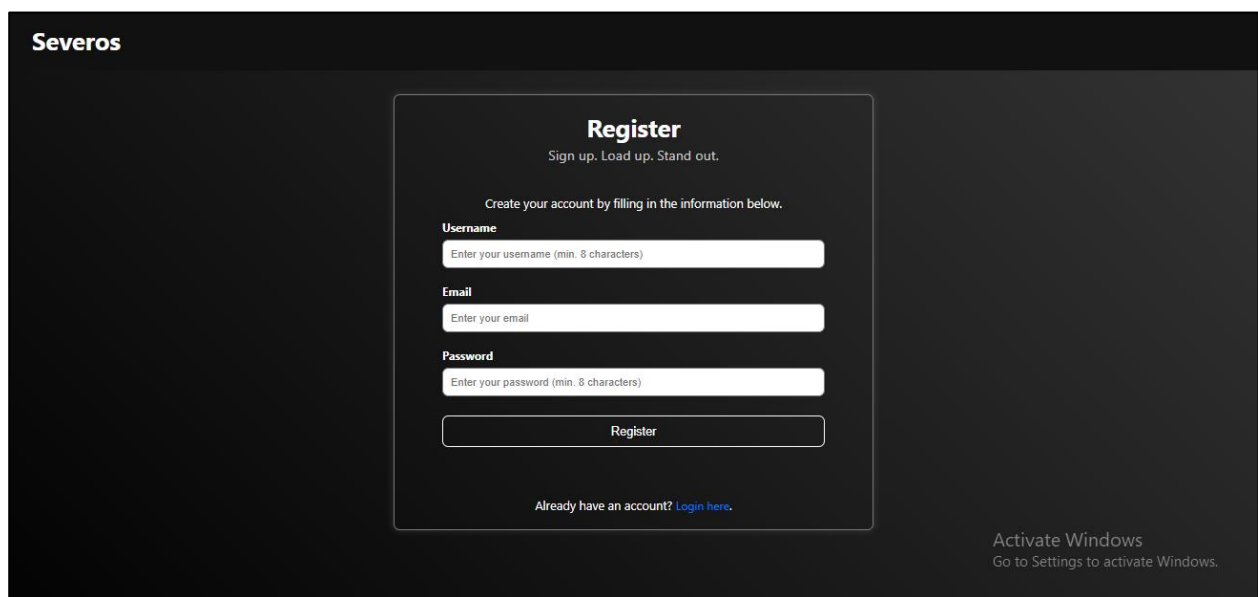


Figure 5. Register Page

- **Validate** that the input must not be empty. If not, **display** error message.
- **Validate** that the username and password length must be **greater than equal to 8** characters.
- **Validate** the email format manually.
- **Store** the user data in the database after successful validation and **hash** the password using **bcrypt hashing**.

4. Marketplace Page

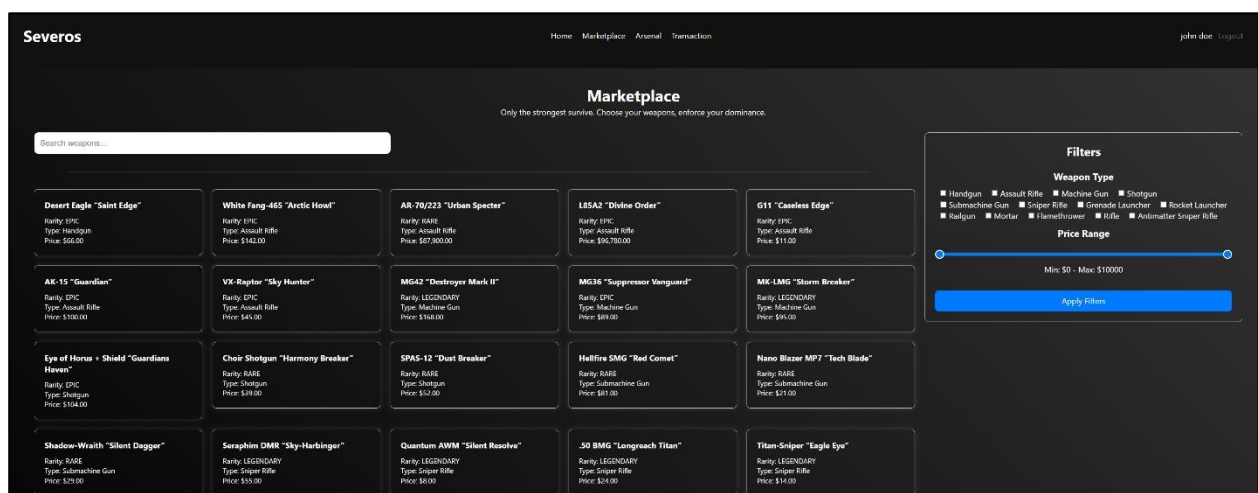


Figure 6. Marketplace Page

- **Add** search bars for users to search for weapon names.

- **Add** filter based on weapon type and price range. **Apply** the filter if user **click** the apply filter button.
- **Display** all weapon that available in the marketplace along with the information such as weapon name, weapon rarity, weapon type, weapon price. **Implement** scrollable to prevent content overflow.
- **Redirect** user to the respective weapon detail page if user click the weapon card.

5. Marketplace Weapon Detail Page

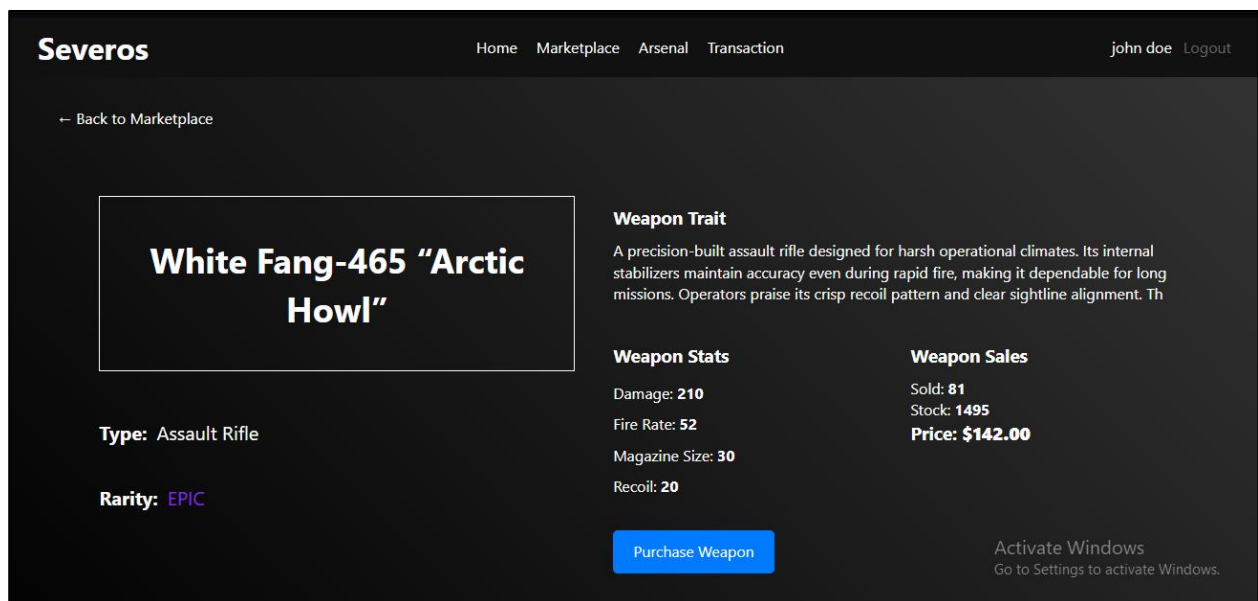


Figure 7. Weapon Detail Page

- **Display** all weapon data such as weapon name, weapon type, weapon rarity, weapon trait, weapon damage, weapon fire rate, weapon magazine size, weapon recoil, weapon sold, weapon stock, and weapon price.
- **Add** back button for user to **go back** to marketplace page.
- **Add** purchase button to **redirect** user to checkout page.

6. Checkout Page

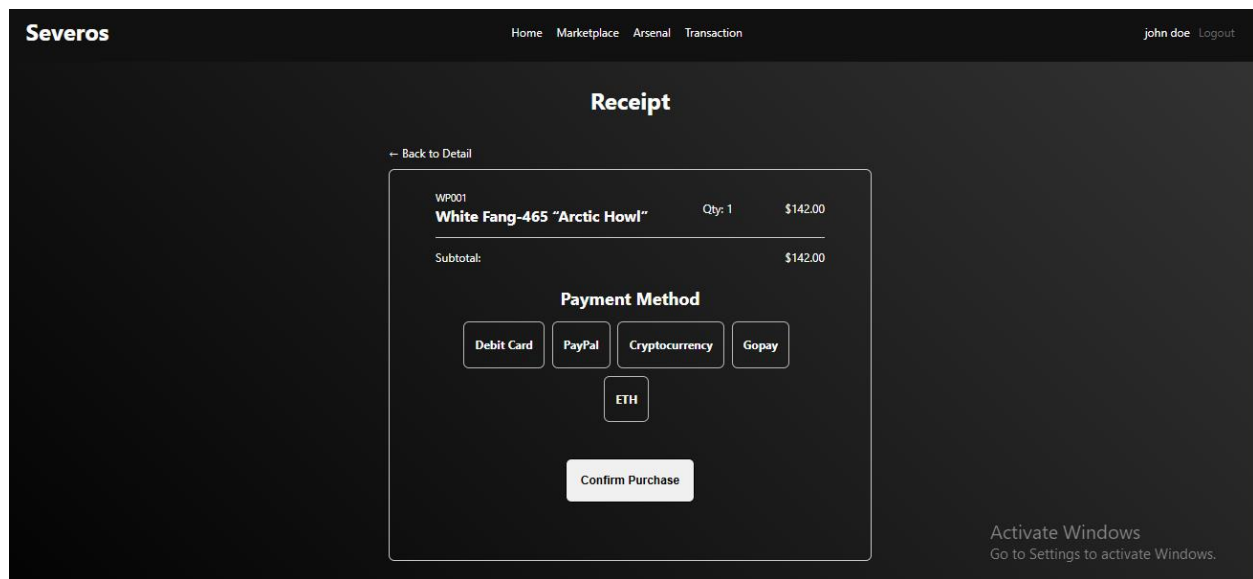


Figure 8. Checkout Page

- **Display** weapon name, weapon id, weapon quantity, and weapon price. Every transaction **can only purchase 1 weapon**.
- **Display** transaction subtotal
- **Display** all available payment methods. **Validate** that user must choose one of the available methods.
- **Add** submit button to confirm purchase. **Redirect** user back to marketplace page after successful purchase.
- **Add** back button to cancel the transaction. **Redirect** user back to weapon detail page.

7. Arsenal Page

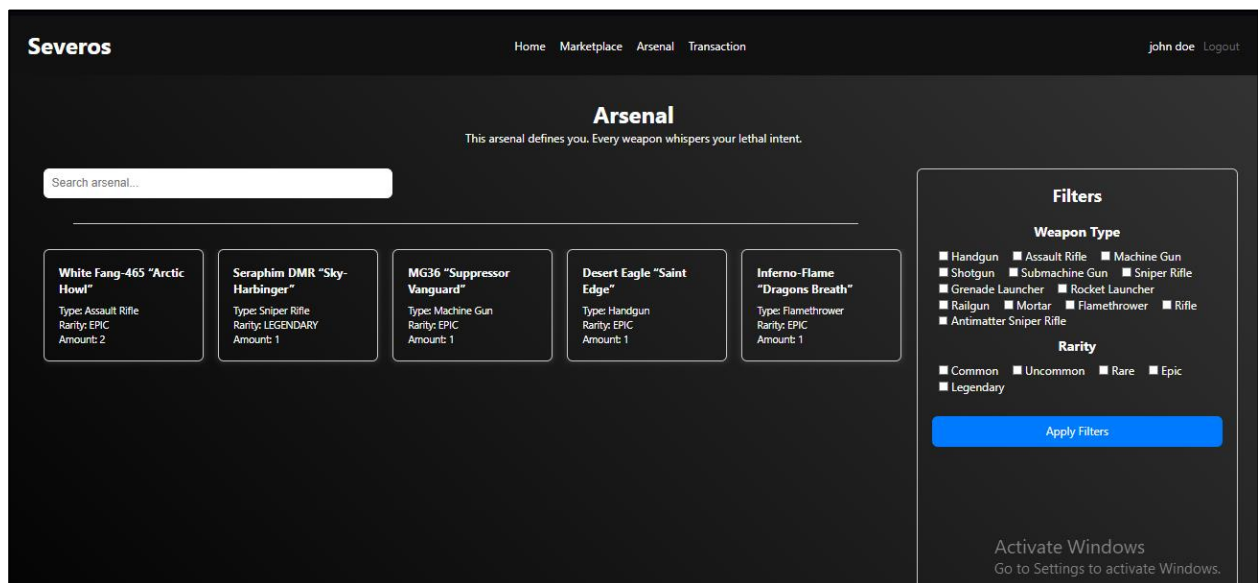


Figure 9. Arsenal Page

- **Add** search bars for users to search for weapon names.
- **Add** filter based on weapon type and weapon rarity. **Apply** the filter if user **click** the apply filter button.
- **Display** all weapon that available in the respective user arsenal along with the information such as weapon name, weapon rarity, weapon type, weapon price. **Implement** scrollable to prevent content overflow.
- **Redirect** user to the respective arsenal weapon detail page if user click the weapon card.

8. Arsenal Weapon Detail Page

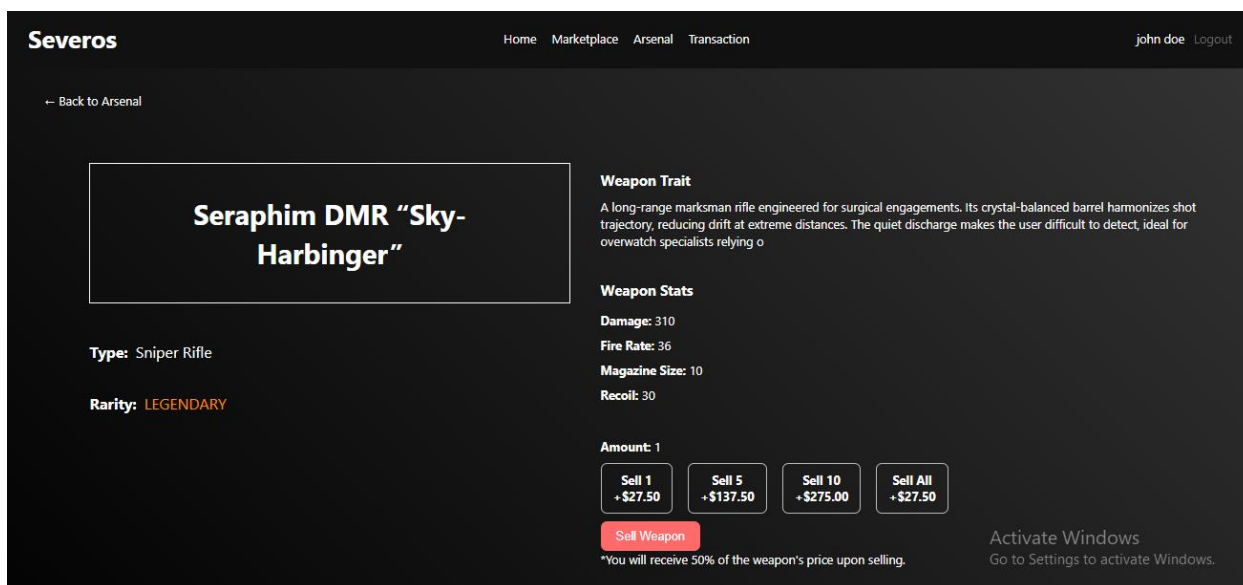


Figure 10. Arsenal Weapon Detail Page

- **Display** all weapon data such as weapon name, weapon type, weapon rarity, weapon trait, weapon damage, weapon fire rate, weapon magazine size, weapon recoil, weapon amount.
- **Add** sell quantity options for users to sell their weapon. Sell compensation is half of the weapon price.

$$\text{Weapon Compensation} = 0.5 \times \text{Weapon Price}$$

Example:

$$\text{Weapon Price} = 98000$$

$$\text{Weapon Compensation} = 0.5 \times 98000 = 49000$$

Figure 11. Weapon Compensation Formula

- **Add** back button for user to **go back** to arsenal page.
- **Add** sell button to sell the weapon.

9. Transaction Page

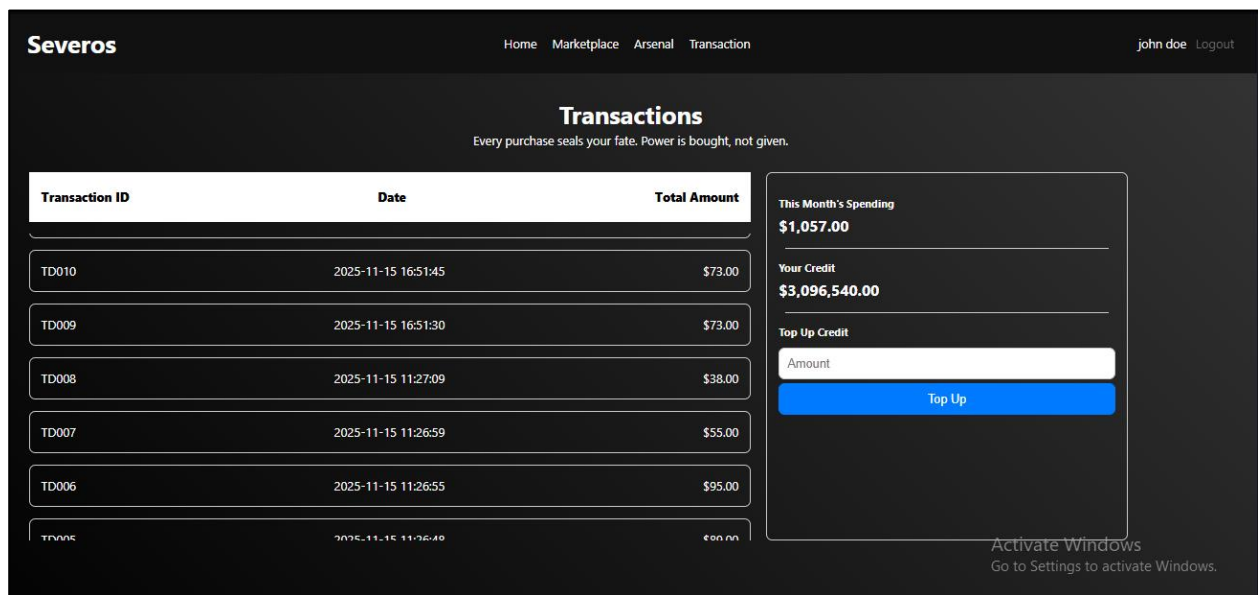


Figure 12. Transaction Page

- **Display** current user transaction history consists of transaction id, transaction date, transaction subtotal. **Implement** scrollable to prevent content overflow.
- **Display** current user balance overview such as this month spending and user credit.
- **Add** top up form for users to recharge their credit balance for transactions. **Validate** that the input must be **positive integer greater than 0**.
- **Add** top button to confirm the top up transaction.

10. Admin Dashboard Page

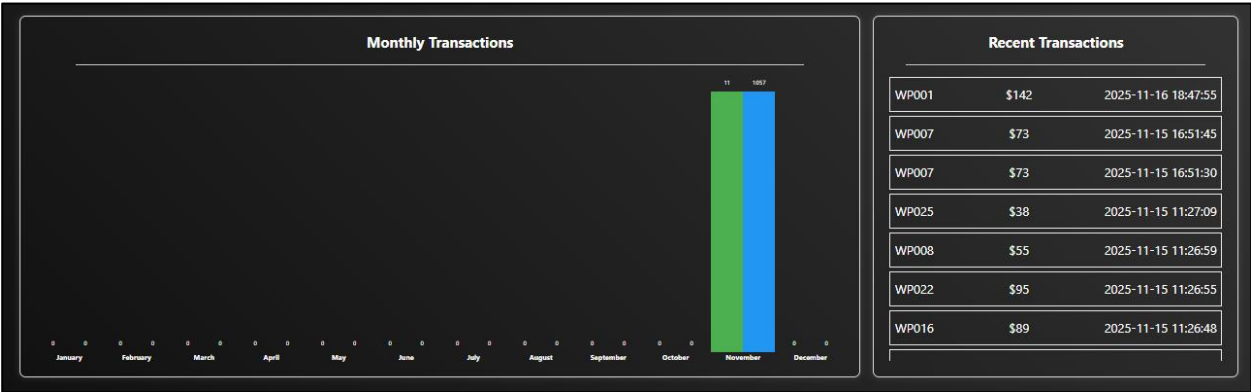


Figure 13. Admin Dashboard Transaction Overview

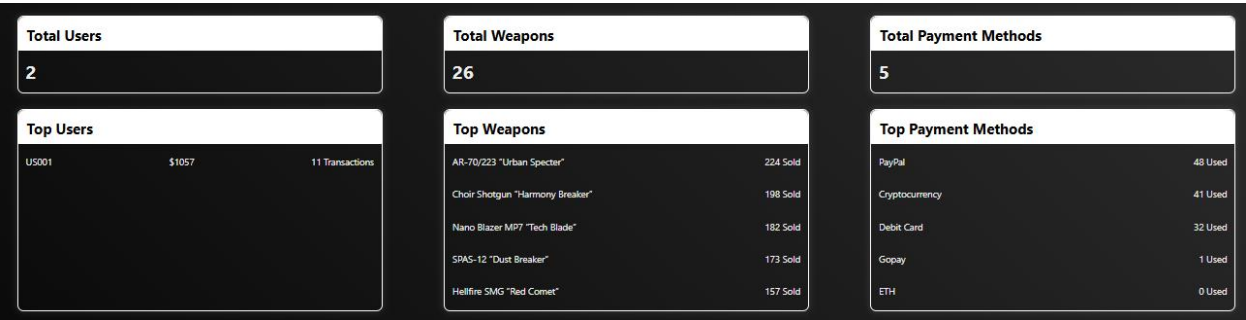


Figure 14. Admin Dashboard User, Weapon, and Payment Overview

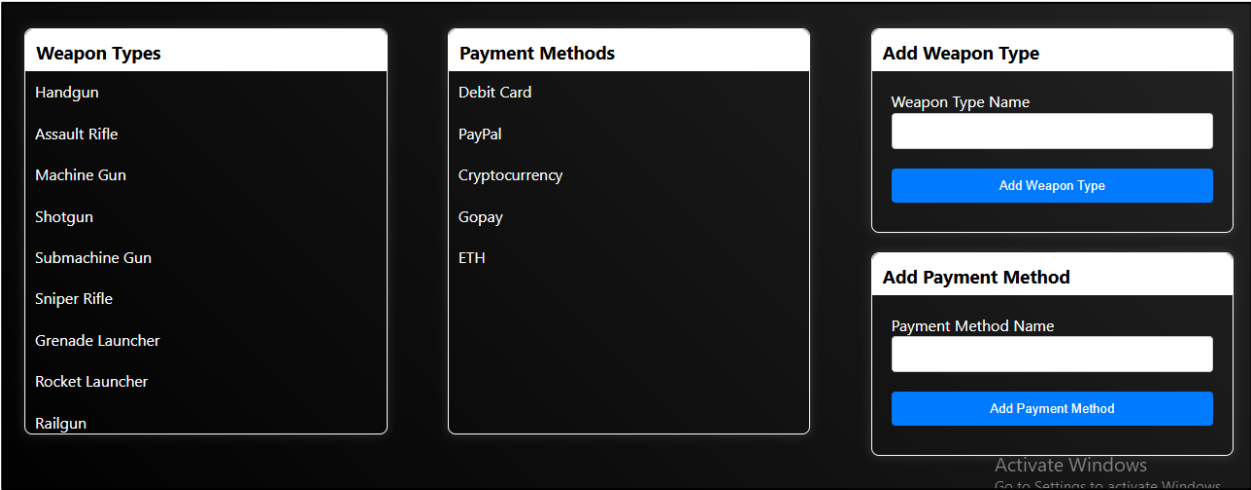


Figure 15. Admin Dashboard Weapon Type and Payment Method

- **Display** transaction overview such as recent transaction and transaction yearly performance. You are free to design the layout as long as it's informative. **Implement** scrollable to prevent content from overflowing.
- **Display** user overview such as total user and top active user.
- **Display** weapon overview such as total weapon and top selling weapon.
- **Display** payment overview such as total payment method and top used payment method.

- **Display** weapon type list that available. **Implement** scrollable to prevent content from overflowing.
- **Display** payment method list that available. **Implement** scrollable to prevent content from overflowing.
- **Add** form for adding new wepaon type and new payment method.

11. Admin Weapon List Page

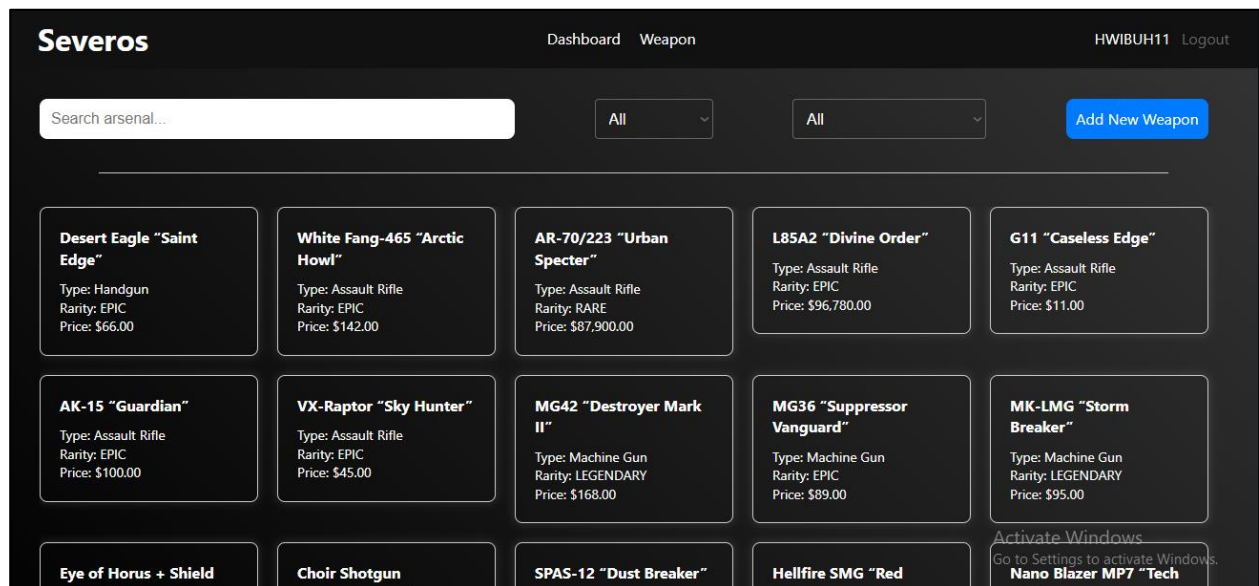


Figure 13. Admin Weapon List Page

- **Add** search bars for users to search for weapon names.
- **Add** dropdown for filter based on weapon type and weapon rarity.
- **Add** add new wepaon button. **Redirect** user to add weapon page.
- **Display** all available weapon list. **Implement** scrollable to prevent content from overflowing. **Redirect** user to respective weapon detail page if user **click** weapon card.

12. Admin Weapon Detail Page

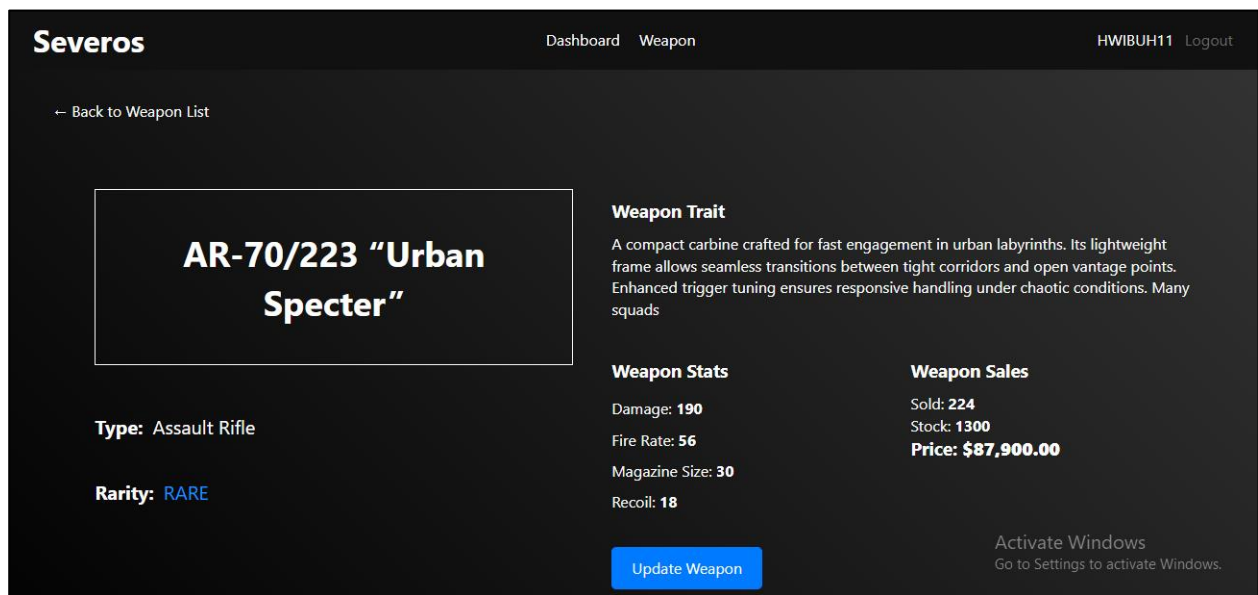


Figure 17. Admin Weapon Detail Page

- **Display** all weapon data such as weapon name, weapon type, weapon rarity, weapon trait, weapon damage, weapon fire rate, weapon magazine size, weapon recoil, weapon sold, weapon stock, and weapon price.
- **Add** back button for user to **go back** to weapon list page.
- **Add** update button to **redirect** user to weapon update page.

13. Admin Update Weapon Page

Severos

DashboardWeapon

HWIBUH11Logout

[← Back to Weapon Detail](#)

AR-70/223 “Urban Spect

Type:Assault Rifle

Rarity:RARE

Weapon Trait

A compact carbine crafted for fast engagement in urban labyrinths. Its lightweight frame allows seamless transitions between tight corridors and open vantage points. Enhanced trigger tuning ensures responsive handling under chaotic conditions.

Weapon Stats

Damage:190

Fire Rate:56

Magazine Size:30

Recoil:18

Weapon Sales

Sold:224

Stock:1300

Price:\$87900

Update Weapon

Activate Windows
Go to Settings to activate Windows.

Figure 18. Admin Update Weapon Page

- **Display** input field for all weapon data such as weapon name, weapon type, weapon rarity, weapon trait, weapon damage, weapon fire rate, weapon magazine size, weapon recoil, weapon stock, and weapon price.
- **Add** button to update the weapon. **Redirect** user back to weapon detail after successful update.
- **Add** back button to **redirect** user back to weapon detail page.

14. Admin Add Weapon Page

Severos

DashboardWeapon

HWIBUH11Logout

[← Back to Weapon List](#)

Type:Select Type

Rarity:Select Rarity

Weapon Trait

Weapon Stats

Damage:0

Fire Rate:0

Magazine Size:0

Recoil:0

Weapon Sales

Sold:0

Stock:0

Price:\$0

Add Weapon

Activate Windows
Go to Settings to activate Windows.

Figure 19. Add Weapon Page

- **Display** input field for all weapon data such as weapon name, weapon type, weapon rarity, weapon trait, weapon damage, weapon fire rate, weapon magazine size, weapon recoil, weapon stock, and weapon price.
- **Add** button to add the weapon. **Redirect** user back to weapon list after successful creation.
- **Add** back button to **redirect** user back to weapon detail page.